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Project Checkpoint Report 1

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Project: ResearchNarrative - RAG-Based Research Storyline Generation Engine

1. Project Plan (Scope):

1.1 Problem Statement:

Academic literature search has fundamentally remained unchanged for decades. Current tools like Google Scholar, Semantic Scholar, and arXiv search return ranked lists of papers based on keyword relevance or citation counts. However, these tools fail to answer critical research questions that scholars actually need:

- *What is the main storyline of this research area?*
- *Which ideas are competing, and how did one approach replace or refine another?*
- *What is the current frontier, and what problems remain open?*

Researchers spend weeks manually reading papers, tracking citations, and mentally constructing the "story" of a field—a cognitively demanding process that current search engines do not support. Our project addresses this gap by developing **ResearchNarrative**, a RAG-based Research Storyline Generation Engine that transforms academic paper collections into structured, temporally-aware narratives with explicit citation relationships, competing approaches, and evolutionary threads.

1.2 Background & Need

- Information Overload Crisis: Over 5 million new research papers are published annually across arXiv, PubMed, IEEE, and ACM alone. Researchers cannot manually synthesize this volume into coherent understanding.

- List-Based Search Paradigm Failure: Existing academic search tools (Google Scholar, Semantic Scholar, arXiv) optimize for retrieval relevance, not narrative understanding. They return isolated papers rather than connected knowledge structures.

- Survey Paper Lag: Traditional survey papers take 1–2 years to write and publish, becoming outdated before release. No automated system exists to generate living, up-to-date research narratives.

- Citation Graph Underutilization: Citation networks encode rich semantic relationships (builds-upon, criticizes, extends, replaces) that current tools display as simple counts rather than storyline elements.

- LLM Hallucination Problem: Direct LLM queries about research fields produce plausible but often factually incorrect summaries. RAG-based approaches with verified academic sources can provide grounded, citation-backed narratives.
- Temporal Dynamics Ignored: Research fields evolve through distinct phases (emergence, competition, consolidation, paradigm shifts). Current tools provide static snapshots rather than dynamic evolutionary views.
- Cross-Disciplinary Synthesis Gap: Important innovations often occur at field intersections, but existing tools are limited by discipline and cannot trace idea transfer across domains.

1.3 Concept

ResearchNarrative transforms literature search from a retrieval task into a **narrative generation task**:

- **Phase 1 – Literature Retrieval:** Automated paper collection from arXiv, Semantic Scholar, and OpenAlex APIs based on user-specified research topics.
- **Phase 2 – Semantic Representation:** Embedding-based representation of papers using Sentence Transformers, enabling semantic similarity computation and clustering.
- **Phase 3 – Thread Discovery:** Unsupervised clustering to identify distinct research threads (e.g., "attention mechanisms," "transformer architectures," "efficient transformers") with LLM-assisted cluster labeling.
- **Phase 4 – Temporal-Citation Analysis:** Graph-based analysis of citation relationships to identify emergence, competition, dominance, and decline patterns within and across threads.
- **Phase 5 – RAG Story Generation:** Retrieval-augmented generation producing structured narratives with explicit paper citations, timeline organization, and competing-approach comparisons.
- **Phase 6 – Interactive Visualization:** Dashboard for exploring research timelines, thread relationships, and navigating the generated narrative with direct paper links

1.4 Point A: Current State & Related Work

We have established the foundational infrastructure for the ResearchNarrative system, including API integrations, data pipelines, and initial prototype components.

Data Sources Established:

- **arXiv API:** Configured for bulk metadata retrieval (title, abstract, authors, categories, dates) with rate limiting and pagination support.

- **Semantic Scholar API:** Integrated for citation graph extraction, paper influence scores, and author disambiguation.

- **OpenAlex API:** Connected for clean metadata, institutional affiliations, and cross-database paper matching.

Literature Review & Novelty:

AI-assisted literature review tools exist but remain limited in scope and narrative capability:

- **Elicit (2023)** provides AI-powered paper discovery and question-answering but produces isolated answers rather than connected narratives [1].
- **ResearchRabbit (2022)** visualizes citation networks but lacks temporal narrative generation and storyline synthesis [2].
- **Consensus (2023)** aggregates findings across papers but does not identify competing approaches or evolutionary patterns [3].
- **Connected Papers (2020)** offers visual citation graphs but provides no narrative explanation of relationships or temporal dynamics [4].
- **Semantic Scholar TLDR (2021)** generates single-paper summaries using BART but does not synthesize across multiple papers into coherent storylines [5].
- **SciSpace/Typeset (2023)** offers paper explanation and Q&A but focuses on individual paper comprehension rather than field-level synthesis [6].

Our Novelty:

ResearchNarrative advances beyond existing tools through:

1. **Thread Discovery:** Automatic identification of distinct research directions within a topic using embedding-based clustering.
2. **Temporal Narrative Structure:** Organization of papers into chronological threads with explicit phase labeling (emergence, peak, decline, replacement).
3. **Competition Modeling:** Citation-graph analysis to identify rival approaches and track dominance shifts over time.
4. **RAG-Grounded Generation:** All narrative claims are backed by specific paper citations, eliminating LLM hallucination risks.
5. **Multi-Source Integration:** Unified pipeline combining arXiv, Semantic Scholar, and OpenAlex for comprehensive coverage.

1.5 Point B: Planned Deliverables

By the end of the semester, we aim to deliver:

- **Paper Ingestion & Retrieval System** - Multi-API integration (arXiv, Semantic Scholar, OpenAlex) with unified schema. - Topic-based paper retrieval with filtering by year, citation count, and venue. Efficient caching and incremental update mechanisms for large paper collections.
- **Embedding & Clustering Pipeline** - Sentence Transformer embeddings (all-MiniLM-L6-v2 / SPECTER) for paper representations. FAISS-based vector storage for efficient similarity search. Hierarchical clustering (HDBSCAN) for research thread discovery. - LLM-assisted cluster labeling generating human-readable thread names.
- **Citation Graph Analysis Module** - Citation network construction from Semantic Scholar data. Temporal influence scoring (papers that caused paradigm shifts). - Competition detection (clusters that cite each other critically). Dominance tracking

(which approaches "won" at different time periods).

- **RAG-Based Narrative Generation** - Structured prompt templates for storyline generation. - Chunk-level retrieval for grounded generation. Citation verification ensuring all claims reference specific papers. Multi-section output: Origins, Competing Approaches, Evolution, Current State, Open Problems.
- **Interactive Dashboard** - Streamlit-based visualization of research timelines. Thread-based navigation with paper drill-down. - Generated narrative display with clickable citations. Export functionality for Markdown/PDF reports.
- **Evaluation Framework** - Factual accuracy assessment (citation verification). Coverage evaluation (important papers included). - Coherence scoring (narrative flow quality). User study with graduate students comparing to manual literature review.

1.6 Milestone Chart

Checkpoint	Planned Work
CP1 – Proposal	Defined project vision and high-level architecture for ResearchNarrative. Completed literature review on AI-assisted literature tools, RAG systems, and academic search. Established API access to arXiv, Semantic Scholar, and OpenAlex. Created initial data models for paper representation.
CP2 – Data Pipeline & Embedding System	Major milestone achieved. Implemented complete paper ingestion pipeline from arXiv and Semantic Scholar APIs with unified schema normalization. Built embedding generation pipeline using Sentence Transformers (SPECTER model). Integrated FAISS vector store for efficient similarity search. Developed HDBSCAN-based clustering for automatic thread discovery. Created Streamlit dashboard prototype displaying clusters, timelines, and sample narrative generation. Achieved end-to-end functionality from topic query → paper retrieval → clustering → basic narrative output.
CP3 – Citation Graph Analysis	Extend pipeline to include full citation graph construction using Semantic Scholar API. Implement temporal influence scoring algorithms. Develop competition detection between research threads. Create visualization for citation flow and dominance patterns.
CP4 – RAG Narrative Engine	Implement production-grade RAG pipeline with structured prompts. Develop multi-section narrative templates (Origins, Competition, Evolution, Frontier). Integrate citation verification ensuring hallucination-free generation. Add narrative chunking for long research areas.
CP5 – Evaluation & Refinement	Conduct factual accuracy evaluation on generated narratives. Perform coverage assessment comparing to expert-curated surveys. Execute user study with 10+ graduate students. Iterate on prompt engineering based on feedback.
CP6 – Final Delivery	Deliver complete ResearchNarrative system with multi-topic support. Finalize dashboard with export functionality. Compile comprehensive documentation and demonstration video. Prepare final report with quantitative evaluation results and case studies.

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2. Current Progress Report (Match):

Work Done (Proposal Phase)

- **Completed comprehensive literature review** on AI-assisted literature tools, RAG systems, and academic search platforms, identifying gaps in existing solutions and establishing project novelty.
- **Defined end-to-end system architecture** covering all six phases: Literature Retrieval, Semantic Representation, Thread Discovery, Temporal-Citation Analysis, RAG Story Generation, and Interactive Visualization.
- **Established API access and documentation review** for primary data sources:
  - arXiv API: Reviewed bulk metadata retrieval capabilities, rate limits, and query syntax.
  - Semantic Scholar API: Obtained API key and reviewed citation graph endpoints.
  - OpenAlex API: Explored metadata schema and cross-database linking capabilities.
- **Created initial data models** defining paper representation schema (title, abstract, authors, year, citations, embeddings) for unified storage across multiple sources.
- **Identified technology stack** for implementation: Python, Sentence Transformers (SPECTER), FAISS, HDBSCAN, LangChain/LlamaIndex for RAG, Streamlit for dashboard.
- **Developed project timeline** with concrete milestones for each checkpoint, ensuring feasibility within semester constraints.

Planned vs Achieved

- **Initially Planned:** Define project scope, complete literature review, establish data source access, and create system architecture.
- **Achieved:** All proposal-phase objectives completed. Comprehensive literature review identifies clear gaps in existing tools. System architecture defined with specific technologies for each component. API access established for all three primary data sources. Milestone chart created with realistic deliverables per checkpoint.

Next 2 Weeks (Upcoming Work)

- **Implement paper ingestion pipeline** with arXiv and Semantic Scholar API clients, including rate limiting, pagination, and error handling.
- **Build unified data normalization layer** to map heterogeneous API responses into consistent paper schema.
- **Set up embedding generation pipeline** using SPECTER model for scientific paper representations.
- **Integrate FAISS vector store** for efficient similarity search and persistence.

- **Begin HDBSCAN clustering experiments** on sample paper collections to validate thread discovery approach.
- **Create initial Streamlit dashboard skeleton** with basic topic search functionality.

Potential Risks & Mitigations

- **API Rate Limits:** arXiv and Semantic Scholar impose rate limits that may slow data collection. *Mitigation:* Implement caching, batch requests efficiently, and pre-collect datasets for target topics.
- **Embedding Quality:** General-purpose embeddings may not capture scientific semantics well. *Mitigation:* Use SPECTER (citation-informed) or SciBERT; evaluate multiple models early.
- **Cluster Interpretability:** Unsupervised clusters may not align with human-recognizable research threads. *Mitigation:* Add LLM-assisted labeling and manual validation on sample topics.
- **Narrative Coherence:** RAG-generated narratives may lack flow or miss important papers. *Mitigation:* Develop structured prompt templates and include citation verification checks.

3. Supporting Evidence (Factual):

Repository and Documentation

Project repository initialized with proposal documentation and architecture designs:

GitHub Repository:
https://github.gatech.edu/bchamarthi6/CS6235_ResearchNarrative.git

The repository currently includes: - Project proposal document with problem statement and motivation, we will be adding code files and other artifacts post the proposal stage..

Key Deliverables Submitted for Checkpoint 1

- **Project Proposal:** Comprehensive document defining problem statement, background, concept, and planned deliverables.
- **Literature Review:** Analysis of 10+ existing tools (Elicit, ResearchRabbit, Connected Papers, etc.) with identified gaps and novelty justification.
- **Data Source Analysis:** exploration of API capabilities, rate limits, and data schemas for arXiv, Semantic Scholar, and OpenAlex.
- **Technology Selection:** Justified choices for embedding models (SPECTER), vector stores (FAISS), clustering (HDBSCAN), and RAG frameworks (LangChain).
- **Milestone Chart:** Detailed checkpoint-by-checkpoint plan with specific deliverables and success criteria.

Feasibility Validation

- **API Access Confirmed:** Successfully obtained API keys and tested basic queries for all three data sources.
- **Technology Compatibility:** Verified that selected libraries (Sentence Transformers, FAISS, HDBSCAN, Streamlit) are compatible and well-documented.
- **Data Availability:** Confirmed that arXiv and Semantic Scholar provide sufficient metadata (titles, abstracts, citations) for planned pipeline.

Validation of Proposal

The repository and submitted documents demonstrate thorough planning and feasibility analysis. The literature review establishes clear differentiation from existing tools, the architecture provides a concrete implementation roadmap, and the milestone chart ensures accountability for measurable progress at each checkpoint.

4. Skill Learning Report:

Retrieval-Augmented Generation (RAG) Fundamentals:

Studied RAG architecture patterns through research papers [9] and documentation. Understood the retrieval-generation pipeline, chunking strategies, and how citation grounding can prevent hallucinations. Identified LangChain and LlamaIndex as candidate frameworks for implementation.

Embedding Models for Scientific Text (SPECTER, SciBERT):

Reviewed literature on domain-specific embedding models, particularly SPECTER [7] which uses citation relationships during training. Understood why general-purpose embeddings may underperform for scientific text and planned comparative evaluation.

Vector Database Concepts (FAISS, pgvector):

Studied FAISS documentation and indexing strategies (Flat, IVF, HNSW). Learned theoretical trade-offs between build time, search speed, and recall accuracy. Identified IVF as initial approach for moderate-scale paper collections.

Clustering Methods for Document Organization:

Reviewed HDBSCAN [8] as primary clustering approach for its ability to handle variable-density clusters and noise. Studied hyperparameter effects (min_cluster_size, min_samples) and evaluation metrics (silhouette score, Davies-Bouldin index).

Academic API Landscape:

Explored documentation for arXiv API, Semantic Scholar Academic Graph API [10], and OpenAlex API. Understood rate limits, query capabilities, and data schemas. Identified strategies for unified normalization across heterogeneous sources.

Graph Analysis for Citation Networks:

Began studying PageRank variants for academic influence scoring, temporal citation analysis, and community detection algorithms. Identified NetworkX and graph-tool as candidate libraries for implementation.

5. Self-Evaluation:

Scope: 110%

- The project addresses a novel problem space (narrative generation vs. paper retrieval) with clear differentiation from existing tools established through comprehensive literature review.
- Architecture covers the complete pipeline from data ingestion through embedding, clustering, citation analysis, to RAG-based narrative generation.
- Scope is ambitious but feasible: includes both technical ML components (embeddings, clustering, RAG) and user-facing deliverables (interactive dashboard, exportable reports).
- Clear novelty established through comparison with 6+ existing tools, identifying specific gaps that ResearchNarrative addresses.

Match: 100%

- All Checkpoint 1 (Proposal) objectives completed as planned.
- Comprehensive literature review with gap analysis completed.
- System architecture fully defined with technology selections justified.
- API access established for all primary data sources.
- Milestone chart created with concrete deliverables for each checkpoint.
- Risk analysis performed with mitigation strategies identified.

Factual: 100%

- Project proposal document submitted with all required sections.
- Literature review references 10+ existing tools and research papers.
- API access confirmed through test queries to arXiv and Semantic Scholar.
- Technology stack documented with justification for each choice.

6. References

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